

Elliptic Curves

The j-function & Uniformization

Definition (j-function)

The modular j -invariant is the function $j(\tau) := 1728 \frac{g_2(\tau)^3}{\Delta(\tau)}$.

We will see in [AEC Proposition I.1.4b. \(j-invariant\)](#) that $j(\tau)$ is the j -invariant of the elliptic curve

$$E_{\Lambda_\tau} : y^2 = 4x^3 - g_2(\tau)x - g_3(\tau),$$

and in [AEC Theorem VI.3.6. \(Tori-curve isomorphism\)](#) that $E_{\Lambda_\tau}(\mathbb{C})$ admits the parametrization

$$\begin{aligned} \mathbb{C}/\Lambda_\tau &\longrightarrow E_{\Lambda_\tau}(\mathbb{C}), \\ z &\longmapsto (\wp(z; \Lambda_\tau), \wp'(z; \Lambda_\tau)). \end{aligned}$$

ATAEC Theorem I.4.1. (Conformal equivalence of modular curve)

$j(\tau)$ is a modular function of weight 0, and induces a biholomorphism $j : X(1) \xrightarrow{\sim} \mathbb{P}^1(\mathbb{C})$.

Proof.

We only verify that j is bijective.

With an appropriate contour argument, one can show that the function $j(\tau) - \tau_0$ has exactly one zero up to multiplicity in the fundamental domain for all $\tau_0 \in \mathbb{C}$.

Also note that $g_2(i\infty) = 120\zeta(4) \neq 0$ and $\Delta(i\infty) = 0$, so j has exactly one simple pole at infinity. ■

ATAEC Corollary I.4.2. (Field of modular functions)

1. If f is a modular function of weight 0, then $f \in \mathbb{C}(j)$.
2. If f is furthermore holomorphic on $\mathbf{H}$, then $f \in \mathbb{C}[j]$.

Proof.

1. By [ATAEC Theorem I.4.1. \(Conformal equivalence of modular curve\)](#), $f \circ j^{-1}$ is a meromorphic function on $\mathbb{P}^1(\mathbb{C})$, such that it is necessarily some rational function $P(t) \in \mathbb{C}(t)$. Substituting $t = j(x)$ directly gives $f(x) = P(j(x))$.
2. Write $f = P(j)$ for some $P(t) \in \mathbb{C}(t)$. If P is not a polynomial, Then there is a $t_0 \in \mathbb{C}$ such that $P(t_0) = \infty$. Then by [ATAEC Theorem I.4.1. \(Conformal equivalence of modular curve\)](#) there exists $\tau_0 \in \mathbf{H}$ with $j(\tau_0) = t_0$. But then $f(\tau_0) = P(j(\tau_0)) = P(t_0) = \infty$, which is a contradiction. ■

Corollary. (Little Picard's Theorem)

Suppose that f is an entire function. Then f omits at least two values in $\mathbb{C}$ iff it is constant.

Proof.

Notice that $j(\tau)$ is a one to one function everywhere on the fundamental domain except for the elliptic points i and ρ (and their $\mathrm{SL}_2(\mathbb{Z})$ -conjugates).

Moreover one can compute that $j(\rho) = 0$ and $j(i) = 1728$.

Hence, by shifting and rescaling if necessary, WLOG suppose that f omits the values 0 and 1728.

Then $j^{-1} \circ f(z)$ is a well-defined holomorphic function from $\mathbb{C}$ to $\mathbb{H}$.

But then $\exp(i \cdot j^{-1} \circ f(z))$ is a bounded entire function, which violates Liouville's theorem. ■

ATAEC Corollary I.4.3. (Uniformization Theorem For Elliptic Curves over $\mathbb{C}$)

Let $A, B \in \mathbb{C}$ satisfy $4A^3 + 27B^2 \neq 0$. Then there is a unique lattice $\Lambda \subset \mathbb{C}$ such that

$$g_2(\Lambda) = 60G_4(\Lambda) = -4A \quad \text{and} \quad g_3(\Lambda) = 140G_6(\Lambda) = -4B$$

so there always exists a corresponding elliptic curve with $y^2 = x^3 + Ax + B$.

- We will see in [AEC Theorem VI.3.6. \(Tori-curve isomorphism\)](#) that this induces a biholomorphism between $\mathbb{C}/\Lambda$ and E .

Proof.

- Existence:

By [ATAEC Theorem I.4.1. \(Conformal equivalence of modular curve\)](#), choose a $\tau \in \mathbb{H}$ such that $j(\tau) = 1728 \frac{4A^3}{4A^3 + 27B^2}$.

There are three cases.

1. $A, B \neq 0$.

Then observe that $\frac{27B^2}{4A^3} = \frac{1728}{j(\tau)} - 1 = -\frac{27g_3(\tau)^2}{g_2(\tau)^3}$, such that $(\frac{B}{g_3(\tau)})^2 (\frac{g_2(\tau)}{A})^3 = -4$.

Let $\alpha = \sqrt{\frac{Ag_3(\tau)}{Bg_2(\tau)}}$ and $\Lambda = \alpha\Lambda_\tau = \mathbb{Z}\alpha\tau + \mathbb{Z}\alpha$. Then we get that

$$\begin{aligned} g_2(\Lambda) &= \alpha^{-4} g_2(\Lambda_\tau) = \frac{B^2 g_2(\tau)^3}{A^2 g_3(\tau)^2} = -4A, \\ g_3(\Lambda) &= \alpha^{-6} g_3(\Lambda_\tau) = \frac{B^3 g_2(\tau)^3}{A^3 g_3(\tau)^2} = -4B \end{aligned}$$

as desired.

2. $A = 0$.

Then $j(\tau) = 0$ and $g_2(\tau) = 0$, so we can take $\Lambda = \alpha\Lambda_\tau$ with $\alpha = \sqrt[6]{\frac{g_3(\tau)}{-4B}}$.

3. $B = 0$.

Then $j(\tau) = 1728$ and $g_3(\tau) = 0$, so we can take $\Lambda = \alpha\Lambda_\tau$ with $\alpha = \sqrt[4]{\frac{g_2(\tau)}{-4A}}$.

- Uniqueness:

Suppose that Λ_1, Λ_2 induce the same g_2, g_3 . Then they necessarily have the same j -

invariant, such that they are the same up to homothety. But then as

$G_{2k}(c\Lambda) = c^{-2k}G_{2k}(\Lambda)$ they must necessarily be the same. ■

Complex Tori & Elliptic Curves

AEC Theorem VI.3.2. (Field of elliptic functions)

Every elliptic function on a lattice Λ can be written as a rational function in $\wp$ and $\wp'$; i.e.

$$\mathbb{C}(\Lambda) = \mathbb{C}(\wp(z), \wp'(z)).$$

AEC Theorem VI.3.5. (Associated elliptic curve)

The Laurent series for $\wp$ at $z = 0$ is $\wp(z) = z^{-2} + \sum_{k=1}^{\infty} (2k+1)G_{2k+1}z^{2k}$.

Additionally, $\wp$ and $\wp'$ satisfy the polynomial relation $[\wp'(z)]^2 = 4[\wp(z)]^3 - 60G_4\wp(z) - 140G_6$.

- Hence, it is standard notation to set $g_2 = 60G_4$ and $g_3 = 140G_6$.

AEC Theorem VI.3.6. (Tori-curve isomorphism)

Fix a lattice $\Lambda \subseteq \mathbb{C}$, with associated quantities g_2, g_3 .

- The polynomial $f(x) = 4x^3 - g_2x - g_3$ has distinct roots. Its discriminant $\Delta = g_2^3 - 27g_3^2$ is nonzero.
- The map $\varphi : \mathbb{C}/\Lambda \rightarrow E \subseteq \mathbb{P}^2(\mathbb{C})$ given by $z \mapsto [\wp(z) : \wp'(z) : 1]$ is a complex analytic isomorphism of complex Lie groups.

Proof.

- Let $\{\omega_1, \omega_2\}$ be a basis for Λ , and let $\omega_3 := \omega_1 + \omega_2$. Then as $\wp'(z)$ is an odd elliptic function we have that $\wp'(\omega_1/2) = 0$. Hence by [AEC Theorem VI.3.5. \(Associated elliptic curve\)](#), $f(x)$ vanishes at each $\wp(\omega_i/2)$, such that it suffices to show that these three values are distinct, such that the polynomial has distinct roots.
To that end, notice that $\wp(z) - \wp(\omega_1/2)$ is even such that it has at least a double zero at $z = \omega_1/2$. But then as its only poles modulo Λ is an order 2 pole at $z = 0$, this is its only zero in $\mathbb{C}/\Lambda$ as the degree of any meromorphic function on a compact Riemann surface must be 0. The same argument shows that $\wp(\omega_1/2), \wp(\omega_2/2), \wp(\omega_3/2)$ are distinct.
- That the image of φ is contained in E follows from [AEC Theorem VI.3.5. \(Associated elliptic curve\)](#).
 - Surjectivity: Let $(x, y) \in E$. Then $\wp(z) - x$ is a non-constant elliptic function such that it necessarily has a zero at some $z = a$. So $\wp'(a)^2 = y^2$; and by replacing a by $-a$ if necessary, we obtain $\wp'(a) = y$ as $\wp'$ is an odd elliptic function. Then $\wp(a) = (x, y)$.
 - Injectivity: Suppose that $\varphi(z_1) = \varphi(z_2)$ such that $\wp(z_1) = \wp(z_2)$ and $\wp'(z_1) = \wp'(z_2)$. There are two cases.
 - $z_1 \not\equiv -z_1 \pmod{\Lambda}$.
Then the function $\wp(z) - \wp(z_1)$ has distinct zeros at z_1 and $-z_1$. But z_2 is also a zero of this function, such that z_2 is equivalent to z_1 or $-z_1$ modulo Λ . Notice

that the latter is not possible, as otherwise $\wp'(z_1) = \wp'(z_2) = \wp'(-z_1) = -\wp'(z_1)$; where the proof of (1) implies that the only zeros of $\wp'$ are at $\omega_1/2, \omega_2/2, \omega_3/2$.

But then all these points satisfy $z_1 \equiv -z_1 \pmod{\Lambda}$.

2. $z_1 \equiv -z_1 \pmod{\Lambda}$.

Then the function $\wp(z) - \wp(z_1)$ has a double zero at z_1 and also vanishes at z_2 .

By the same reasoning in (1) we get that $z_1 \equiv z_2 \pmod{\Lambda}$.

- Analytic isomorphism: We compute its effect on the cotangent space. At every point of E , dx/y is a non-vanishing holomorphic differential. Since $\varphi^*\left(\frac{dx}{y}\right) = \frac{d\wp(z)}{\wp'(z)} = dz$ is similarly non-vanishing and holomorphic at every point of $\mathbb{C}/\Lambda$, we see that φ is a local isomorphism by the Inverse Function Theorem. But then φ is bijective so this implies that it is a global isomorphism.
- Group homomorphism: Let $z_1, z_2 \in \mathbb{C}$. Then by [Divisors and Holomorphic Line Bundles# \(Divisors over elliptic curves\)](#) there exists an $f \in \mathbb{C}(\Lambda)$ such that $\text{div}(f) = [z_1 + z_2] - [z_1] - [z_2] + [0]$. By [AEC Theorem VI.3.2. \(Field of elliptic functions\)](#), we can write $f(z) = F(\wp(z), \wp'(z))$ for some $F(X, Y) \in \mathbb{C}(X, Y)$. If F is then considered as an element of $\mathbb{C}(x, y) = \mathbb{C}(E)$, $\text{div}(F) = [\varphi(z_1 + z_2)] - [\varphi(z_1)] - [\varphi(z_2)] + [\varphi(0)]$; where $[\varphi(0)] = [O]$.
To complete the proof, we show that $\text{div}(F) = [O]$:
By [Divisors and Holomorphic Line Bundles > Hartshorne 6.10.2. \(Group law on elliptic curves\)](#), there is a natural association between points on E and elements of $\text{Cl}(E)$ given by $P \mapsto [P] - [O]$. This extends to give us that for every $D \in \text{Div}(E)$ there is a unique associated point P such that $D \sim [P] - [O]$. In particular, if D is principal, then $0 = D \sim [O] - [O]$. ■

AEC Theorem VI.4.1. (Homothety-Holomorphism-Isogeny correspondence)

Suppose that Λ_1, Λ_2 are lattices in $\mathbb{C}$.

1. There exists a bijection $\{\alpha \in \mathbb{C} \mid \alpha\Lambda_1 \subseteq \Lambda_2\} \longleftrightarrow \{\phi : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2 \mid \phi \text{ holomorphic, } \phi(0) = 0\}$ given by $\alpha \mapsto \phi_\alpha$.
2. If E_1, E_2 are elliptic curves given by the lattices Λ_1, Λ_2 , then the natural inclusion $\{\text{isogenies } \phi : E_1 \rightarrow E_2\} \hookrightarrow \{\phi : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2 \mid \phi \text{ holomorphic, } \phi(0) = 0\}$ is a bijection; where an isogeny of elliptic curves is a morphism of the underlying variety that preserves the point at infinity O (or more generally the specified K -rational point).
As a corollary, E_1 and E_2 are isomorphic over $\mathbb{C}$ if and only if Λ_1, Λ_2 are homothetic.
Moreover, every complex analytic map that preserves O is necessarily a homomorphism.

Proof.

1. If $\phi_\alpha = \phi_\beta$ then for all $z \in \mathbb{C}$, $\alpha z = \beta z \pmod{\Lambda_2}$. Hence the map $z \mapsto (\alpha - \beta)z$ sends $\mathbb{C}$ to Λ_2 ; and since Λ_2 is discrete this map must be constant. Therefore $\alpha = \beta$.
Next let $\phi : \mathbb{C}/\Lambda_1 \rightarrow \mathbb{C}/\Lambda_2$ be a holomorphic map with $\phi(0) = 0$. Then, since $\mathbb{C}$ is simply

connected, we can lift ϕ to a holomorphic map $f : \mathbb{C} \rightarrow \mathbb{C}$ with $f(0) = 0$ such that the map commutes with the quotient map.

Now for any $\omega \in \Lambda_1$, $f(z + \omega) = f(z) \pmod{\Lambda_2}$. Again using the discreteness of Λ_1 , we see that $f(z + \omega) - f(z)$ must be independent of z . Thus $f'(z + \omega) = f'(z)$ for all $z \in \mathbb{C}$ and all $\omega \in \Lambda_1$. So $f'(z)$ is a holomorphic elliptic function such that it is constant.

Therefore $f(z) = \alpha z$ as $f(0) = 0$. Moreover as $f(\Lambda_1) \subseteq \Lambda_2$ implies $\alpha\Lambda_1 \subseteq \Lambda_2$, i.e. $\phi = \phi_\alpha$.

2. First note that since an isogeny is given by local sections of global rational functions, the map induced on the corresponding complex tori will be holomorphic. Thus our association $\text{Hom}(E_1, E_2) \rightarrow \text{Holo}(\mathbb{C}/\Lambda_1, \mathbb{C}/\Lambda_2)$ is well-defined and clearly injective.

We now prove surjectivity. From (a), it suffices to consider a map of the form ϕ_α . The induced map $E_1 \rightarrow E_2$ on Weierstrass equations is given by

$$(\wp(z, \Lambda_1) : \wp'(z, \Lambda_1) : 1) \mapsto (\wp(\alpha z, \Lambda_2) : \wp'(\alpha z, \Lambda_2) : 1).$$

So we must show that $\wp(\alpha z, \Lambda_2)$ and $\wp'(\alpha z, \Lambda_2)$ can be expressed as rational functions of $\wp(z, \Lambda_1)$ and $\wp'(z, \Lambda_1)$. But as $\alpha\Lambda_1 \subseteq \Lambda_2$, we have

$$\wp(\alpha(z + \omega), \Lambda_2) = \wp(\alpha z + \alpha\omega, \Lambda_2) = \wp(\alpha z, \Lambda_2) \text{ for all } \omega \in \Lambda_1; \text{ and similarly for } \wp'(\alpha z, \Lambda_2).$$

Thus $\wp(\alpha z, \Lambda_2)$ and $\wp'(\alpha z, \Lambda_2)$ are in $\mathbb{C}(\Lambda_1)$. But then every Λ_1 -elliptic function can be written as a rational function in terms of $\wp(z, \Lambda_1)$ and $\wp'(z, \Lambda_1)$, which gives the desired result. ■

AEC Theorem VI.5.3. (Equivalence of categories)

The following categories are equivalent:

1. Objects: Elliptic curves over $\mathbb{C}$.
Maps: Isogenies.
2. Objects: Elliptic curves over $\mathbb{C}$.
Maps: Complex analytic maps taking O to O .
3. Objects: Lattices $\Lambda \subseteq \mathbb{C}$ up to homothety.
Maps: $\text{Hom}(\Lambda_1, \Lambda_2) := \{\alpha \in \mathbb{C} \mid \alpha\Lambda_1 \subseteq \Lambda_2\}$.

AEC Theorem VI.5.4. (Elements of order m)

Let $E/\mathbb{C}$ be an elliptic curve and $m \in \mathbb{Z}^+$.

1. As abstract groups, $E[m] \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, where $E[m]$ denotes the set of points of E with order m .
2. The multiplication-by- m map $[m] : E \rightarrow E$ has degree m^2 .

Proof.

1. By [AEC Corollary VI.5.1.1.](#), $E(\mathbb{C})$ is isomorphic to some $\mathbb{C}/\Lambda$. Hence $E[m] \cong (\mathbb{C}/\Lambda)[m] \cong \frac{1}{m}\Lambda/\Lambda \cong (\mathbb{Z}/m\mathbb{Z})^2$.
2. As $\text{char } \mathbb{C} = 0$ and $[m]$ is unramified, the degree of $[m]$ is just the number of points in $E[m] = [m]^{-1}(O)$. ■

Definition (Field of moduli)

Let $\{E\}$ denote an isomorphism class of elliptic curves over $\mathbb{C}$.

- We say that $K \subseteq \mathbb{C}$ is a field of definition for $\{E\}$ if there is $E_0 \in \{E\}$ such that E_0 is defined over K .
- We say that K is a field of moduli for $\{E\}$ if $E^\sigma \in \{E\} \iff \sigma|_K = \text{id}$ for all $\sigma \in \text{Aut}(\mathbb{C}/\mathbb{Q})$.

Note that the field of moduli exists and is unique as this implies that K is the fixed field of the group $\{\sigma \in \text{Aut}(\mathbb{C}/\mathbb{Q}) : E^\sigma \in \{E\}\}$; where one understands E^σ to just be the elliptic curve given by σ acting on the coefficients of E .

From the complex analytic viewpoint described above, it is not clear that $j(\{E\})$ should have any relationship to the field of definition and moduli. We see that, surprisingly enough, the j -invariant of a curve encodes a staggering amount of algebraic information.

AEC Proposition III.1.4b. (j-invariant)

The j -invariant $j(E)$ of the elliptic curve $E : y^2 = 4x^3 - g_2(\tau)x - g_3(\tau)$ is

$$j(E) := j(\tau_E) = 1728 \frac{g_2(\tau_E)^3}{g_3(\tau_E)^3 - 27g_3(\tau_E)^2} = j(\tau_E).$$

(The above is known as the Weierstrass form of the elliptic curve.)

- The j -invariant can be defined algebraically for elliptic curves that are not in Weierstrass form, and it is invariant under isomorphism, but we do not discuss that here.
- Two elliptic curves are isomorphic over $\mathbb{C}$ (as varieties) if and only if they have the same j -invariant.

Proof.

By [AEC Theorem VI.4.1. \(Homothety-Holomorphism-Isogeny correspondence\)](#), E_1, E_2 are isomorphic (up to isogeny) if and only if Λ_1, Λ_2 are homothetic. But then the j function is a weight 0 modular function such that it is invariant up to homothety. ■

AEC Proposition III.1.4.c

For any $\{E\}$, there always exists an $E_0 \in \{E\}$ such that E_0 is defined over $\mathbb{Q}(j(\{E\}))$.

Proof.

There are three cases.

1. $j(\{E\}) \neq 0, 1728$.

Then we let $E_0 : y^2 + xy = x^3 - \frac{36}{j_0 - 1728}x - \frac{1}{j_0 - 1728}$.

One can compute that $\Delta = \frac{j_0^2}{(j_0 - 1728)^3}$ and $j(E_0) = j_0$ so we are done.

2. $j(\{E\}) = 0$.

Then we let $E_0 : y^2 + y = x^3$.

One can compute that $\Delta = -27$ and $j(E_0) = 0$ so we are done.

3. $j(\{E\}) = 1728$.

Then we let $E_0 : y^2 = x^3 + x$.

One can compute that $\Delta = -64$ and $j(E_0) = 1728$ so we are done. ■

ATAEC Proposition VI.4.5. (Minimal field of definition)

1. $\mathbb{Q}(j(\{E\}))$ is the field of moduli for $\{E\}$.
2. $\mathbb{Q}(j(\{E\}))$ is the minimal field of definition for $\{E\}$.

Proof.

1. [AEC Proposition I.1.4b. \(j-invariant\)](#) implies that $E^\alpha \in \{E\}$ if and only if $j(E^\sigma) = j(E)$. But then $j(E^\sigma) = j(E)^\sigma$.
2. By [AEC Theorem VI.4.1. \(Homothety-Holomorphism-Isogeny correspondence\)](#), two elliptic curves are isomorphic iff their lattices are homothetic. But then the j -function is invariant under $\text{SL}_2(\mathbb{Z})$, so two elliptic curves are isomorphic iff they have the same j -invariant. As j is built out of the coefficients of the Weierstrass form of E , that $j(\{E\})$ is contained any field of definition for $\{E\}$. By [AEC Proposition III.1.4.c](#) we are done. ■

Complex Multiplication

Definition. (Endomorphism ring)

Let $E/\mathbb{C}$ be an elliptic curve. Notice that (4.1) allows us to identify $\text{End}(E)$ with a certain subring of $\mathbb{C}$. Thus if $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$ as in (5.1.1), then $\text{End}(E) \cong \{x \in \mathbb{C} : x\Lambda \subseteq \Lambda\}$.

- Since Λ is unique up to homothety, this ring is independent of Λ . We now use this description of $\text{End}(E)$ to completely characterize the possible endomorphism rings which can occur.

AEC Theorem VI.5.5. (Classification of endomorphism ring)

Let $E/\mathbb{C}$ be an elliptic curve, and let ω_1, ω_2 be generators for the lattice Λ associated to E . Either

1. $\text{End}(E) = \mathbb{Z}$; or
2. $\mathbb{Q}(\omega_1/\omega_2)$ is a quadratic imaginary extension of $\mathbb{Q}$, and $\text{End}(E)$ is isomorphic to an order in $\mathbb{Q}(\omega_1/\omega_2)$; which is a subring of $\mathbb{Q}(\omega_1/\omega_2)$ that is finitely generated as a $\mathbb{Z}$ -module with $R \otimes \mathbb{Q} = \mathbb{Q}(\omega_1/\omega_2)$.
Hence, if $\text{End}(E) \simeq R$, we say that E has complex multiplication by R or by K ; where $K = R \otimes \mathbb{Q}$.

Proof.

Let $\tau = \omega_1/\omega_2$, such that we may replace Λ by $\Lambda_\tau = \mathbb{Z} + \mathbb{Z}\tau$. Let $R = \{x \in \mathbb{C} : x\Lambda \subseteq \Lambda\}$, such

that $R \cong \text{End}(E)$ by [AEC Theorem VI.4.1.](#) Then for any $\alpha \in R$ we can write $\alpha = a + b\tau$ and $x\tau = c + d\tau$. From this we get that $\alpha^2 - (a + d)\alpha + bc = 0$, so R is an integral extension of $\mathbb{Z}$.

Now suppose that $R \neq \mathbb{Z}$, such that $\alpha \in R \setminus \mathbb{Z}$. Then we can eliminate the system above with respect to τ to get that $b\tau^2 - (a - d)\tau + c = 0$; such that $\mathbb{Q}(\tau)$ is an imaginary quadratic field. Finally as $R \subseteq \mathbb{Q}(\tau)$ and R is integral over $\mathbb{Z}$, it follows that R is an order of $\mathbb{Q}(\tau)$. ■

Two examples.

- $\Lambda = \mathbb{Z}[i]$ is associated with the curve $y^2 = x^3 + x$, and multiplication by i on Λ corresponds to the endomorphism $(x, y) \mapsto (-x, iy)$.
- $\Lambda = \mathbb{Z}[\omega]$ corresponds to the curve $y^2 = x^3 + 1$, and multiplication by ω corresponds to the endomorphism $(x, y) \mapsto (\omega x, -y)$.

Remark.

We have seen that in order to understand particular elliptic curves, it is often useful to study the set (or moduli space!) of all elliptic curves. This motivates us to consider the following set:

$$\mathcal{ELL}(R) := \frac{\{\text{elliptic curves } E/\mathbb{C} \text{ with } \text{End}(E) \cong R\}}{\text{isomorphism over } \mathbb{C}} = \frac{\{\text{lattices } \Lambda \text{ with } \text{End}(E_\Lambda) \cong R\}}{\text{homothety}}.$$

Given a quadratic imaginary field K , how might we construct elements of $\mathcal{ELL}(\mathcal{O}_K)$?

Recall from [Math 204A > Corollary 9.2.](#) that every fractional ideal of K embeds as a lattice in $\mathbb{C}$. Then notice that $\text{End}(\bar{E}_a) = \{\alpha \in \mathbb{C} : \alpha a \subset a\} = \{\alpha \in K : \alpha a \subset a\} \cong \mathcal{O}_K$ as desired.

On the other hand, [AEC Theorem VI.4.1. \(Homothety-Holomorphism-Isogeny correspondence\)](#) tells us that we would only want to consider the a 's which are the same up to homothety; i.e. that we look at the group of fractional ideals modulo principal ideals, which everyone should recognize as the ideal class group $\text{Cl}(K)$.

ATAEC Proposition II.1.2. (Action of ideal class group)

1. Let Λ be a lattice with $E_\Lambda \in \mathcal{ELL}(\mathcal{O}_K)$, and let a and b be non-zero fractional ideals of K .

Then

- $a\Lambda$ is a lattice in $\mathbb{C}$;
- The elliptic curve $E_{a\Lambda}$ satisfies $\text{End}(E_{a\Lambda}) \cong \mathcal{O}_K$;
- $E_{a\Lambda} \cong E_{b\Lambda}$ if and only if $\bar{a} = \bar{b}$ in $\text{Cl}(\mathcal{O}_K)$.

Hence there is a well-defined action of $\text{Cl}(\mathcal{O}_K)$ on $\mathcal{ELL}(\mathcal{O}_K)$ determined by

$$\bar{a} * E_\Lambda = E_{a^{-1}\Lambda}.$$

2. The action of $\text{Cl}(\mathcal{O}_K)$ on $\mathcal{ELL}(\mathcal{O}_K)$ described in (a) is simply transitive; i.e. there is a unique action that carries one specified element to another. In particular, $\#\text{Cl}(\mathcal{O}_K) = \#\mathcal{ELL}(\mathcal{O}_K)$.

Proof.

1. • By assumption, $\text{End}(E_\Lambda) = \mathcal{O}_K$ such that $\mathcal{O}_K \Lambda = \Lambda$. Let d be a big enough integer such that $d\alpha \subset \mathcal{O}_K$ such that $\alpha\Lambda \subset \frac{1}{d}\Lambda$; so $\alpha\Lambda$ is a discrete subgroup of $\mathbb{C}$. On the other hand, there also has to exist some integer d' such that $d'\mathcal{O}_K \subset \alpha$ and $d\Lambda \subset \alpha\Lambda$. Hence $\alpha\Lambda$ spans $\mathbb{C}$ and is a lattice.
 - For any $\alpha \in \mathbb{C}, \alpha \neq 0$, we have $\alpha\alpha\Lambda \subset \alpha\Lambda \iff \alpha^{-1}\alpha\alpha\Lambda \subset \alpha^{-1}\alpha\Lambda \iff \alpha\Lambda \subset \Lambda$. Hence $\text{End}(E_{\alpha\Lambda}) = \text{End}(E_\Lambda) = \mathcal{O}_K$.
 - By [AEC Theorem VI.4.1. \(Homothety-Holomorphism-Isogeny correspondence\)](#) we know that the isomorphism class of $E_{\alpha\Lambda}$ is exactly determined by the homothety class of $\alpha\Lambda$. Hence $E_{\alpha\Lambda} \cong E_{b\Lambda} \iff \Lambda = c\alpha^{-1}b\Lambda \iff \Lambda = c^{-1}\alpha b^{-1}\Lambda$ for some $c \in \mathbb{C}^*$; such that $c\alpha^{-1}b, c^{-1}\alpha b^{-1} \subseteq R_K$ and is necessarily equal to R_K . Hence $\alpha = cb$.
2. Let $E_{\Lambda_1}, E_{\Lambda_2} \in \mathcal{ELL}(\mathcal{O}_K)$. It is sufficient to find an α such that $\bar{\alpha} * E_{\Lambda_1} = E_{\Lambda_2}$. Choose any $\lambda_1 \in \Lambda_1 \setminus \{0\}$ and consider the lattice $\alpha_1 = \frac{1}{\lambda_1}\Lambda_1$. From [AEC Theorem VI.5.5. \(Classification of endomorphism ring\)](#) we have $\alpha_1 \subseteq K$, such that it is a fractional ideal as it is a finitely generated $\mathcal{O}_K$ -module. Similarly choosing a $\lambda_2 \in \Lambda_2 \setminus \{0\}$ we obtain a second fractional ideal $\alpha_2 = \frac{1}{\lambda_2}\Lambda_2$ of K ; such that $\frac{\lambda_2}{\lambda_1}\alpha_2\alpha_1^{-1}\Lambda_1 = \Lambda_2$. Let $\alpha = \alpha_2^{-1}\alpha_1$, such that $\bar{\alpha} * E_{\Lambda_1} = E_{\alpha^{-1}\Lambda_1} = E_{\frac{\lambda_2}{\lambda_1}\Lambda_2} \cong E_{\Lambda_2}$. The fact that it is simply transitive follows as an immediate consequence of the third statement of (1). ■

ATAEC Proposition II.2.1. (Algebraicness of j-invariant)

1. Let $E/\mathbb{C}$ be an elliptic curve, and let $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ be any field automorphism of $\mathbb{C}$. Then $\text{End}(E^\sigma) \cong \text{End}(E)$.
 2. Let $E/\mathbb{C}$ be an elliptic curve with complex multiplication by the ring of integers $\mathcal{O}_K$ of a quadratic imaginary field K . Then $j(E) \in \bar{\mathbb{Q}}$.
 3. $\mathcal{ELL}(\mathcal{O}_K) \cong \frac{\{\text{elliptic curves } E/\mathbb{Q} \text{ with } \text{End}(E) \cong \mathcal{O}_K\}}{\text{isomorphism over } \bar{\mathbb{Q}}}$.
- It turns out that for elliptic curves with complex multiplication, $j(E)$ is in fact an algebraic integer. The proof of this is outside the scope of this talk, so refer to section II.6 of the text for details.

Proof.

1. Clear; as if $\phi : E \rightarrow E$ is an endomorphism of E , then $\phi^\sigma : E^\sigma \rightarrow E^\sigma$ is an endomorphism of E^σ .
2. We have that $j(E^\sigma) = j(E)^\sigma$ and $\text{End}(E^\sigma) \cong \mathcal{O}_K$ by (1). But then [ATAEC Proposition II.1.2. \(Action of ideal class group\)](#) implies that the isomorphism classes of the orbit of E is a finite set (namely the class group). As the j -invariant is invariant under isomorphism, $j(E)^\sigma$ has a finite orbit and must be an algebraic number.
3. Let $\mathcal{ELL}_{\bar{\mathbb{Q}}} := \frac{\{\text{elliptic curves } E/\mathbb{Q} \text{ with } \text{End}(E) \cong \mathcal{O}_K\}}{\text{isomorphism over } \bar{\mathbb{Q}}}$.

We fix some embedding $\bar{\mathbb{Q}} \hookrightarrow \mathbb{C}$, such that there is a naturally induced map

$\varepsilon : \mathcal{ELL}_{\bar{\mathbb{Q}}}(\mathcal{O}_K) \rightarrow \mathcal{ELL}_{\mathbb{C}}(\mathcal{O}_K)$. We show that this map is a bijection.

- Surjectivity: Let $E/\mathbb{C}$ represent an element of $\mathcal{ELL}_{\mathbb{C}}(R_K)$. Then we have:

1. $j(E) \in \bar{\mathbb{Q}}$, from (2);

2. there is an elliptic curve $E'/\mathbb{Q}(j(E))$ with $j(E') = j(E)$, from [AEC Proposition III.1.4.c](#);

3. E' is isomorphic to E over $\mathbb{C}$.

These three facts imply that $\varepsilon(E') = E$, which proves that ε is surjective.

- Injectivity: Next let $E_1/\mathbb{Q}$ and $E_2/\mathbb{Q}$ represent elements of $\mathcal{ELL}_{\overline{\mathbb{Q}}}(\mathcal{R}_K)$, and suppose that $\varepsilon(E_1) = \varepsilon(E_2)$. Then by $j(E_1) = j(E_2)$ by [AEC Proposition III.1.4b. \(j-invariant\)](#), such that they are isomorphic over $\overline{\mathbb{Q}}$ (this is an algebraic fact that we invoke without proof - see the full statement of Proposition III.1.4b in the book). Hence E_1 and E_2 represent the same element of $\mathcal{ELL}_{\mathbb{Q}}(\mathcal{R}_K)$, which shows that ε is also injective. ■

ATAEC Remark II.2.1. (Class number inequality)

By applying (2) of [ATAEC Proposition II.1.2. \(Action of ideal class group\)](#) to (2) of [ATAEC Proposition II.2.1. \(Algebraicity of j-invariant\)](#), we get that $[\mathbb{Q}(j(E)) : \mathbb{Q}] \leq h_K$.

- It turns out that this is in fact an equality.

Remark. (Ramanujan's constant)

We are now well-positioned to explain the phenomenon

$$e^{\pi\sqrt{163}} = 262537412640768743.9999999999999999250072597.$$

It turns out that when $K = \mathbb{Q}(\sqrt{-163})$, $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{-163}}{2}\right]$ is a UFD (i.e. $h_K = 1$).

This lattice corresponds to the point $\tau = \frac{1+\sqrt{-163}}{2}$.

On the other hand, the q -expansion of $j(\tau)$ is given by

$$j(q) = q^{-1} + 744 + 196844q + 21493760q^2 + \dots$$

such that substitution by $q_0 = e^{2\pi i\tau} = -e^{-\pi\sqrt{163}} \approx -3.089 \cdot 10^{-18}$ gives us that $e^{\pi\sqrt{163}}$ differs from an integer by a polynomial in q_0 .